Gianluca Hotz

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# AlwaysOn in SQL Server 2017

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# Who am I?

- Gianluca Hotz | @glhotz | ghotz@ugiss.org
- Independent Consultant, Founder and Mentor SolidQ
  - 20+ years on SQL Server (from 4.21 in 1996)
  - Database modeling and development, sizing and administration, upgrade and migration, performance tuning
- Interests
  - Relational model, DBMS architecture, Security, High Availability and Disaster Recovery
- Community
  - 20 years Microsoft MVP SQL Server (from 1998)
  - Founder and President UGISS
    - User Group Italiano SQL Server (PASS Chapter)

# Whats is AlwaysOn?

- New name for HA and DR technologies
  - Failover Cluster Instance (FCI)
  - Availability Groups (AG)
- Previously named HADRON
  - Keep in mind when searching articles
  - Still present in some DMVs (_hadr_)

# AlwaysOn FCI

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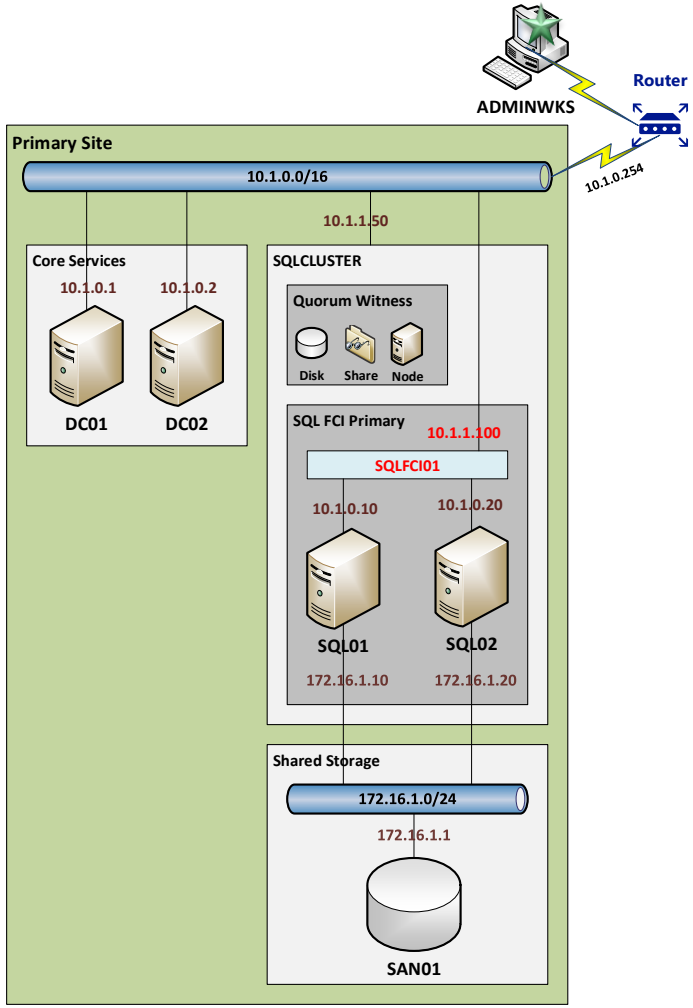
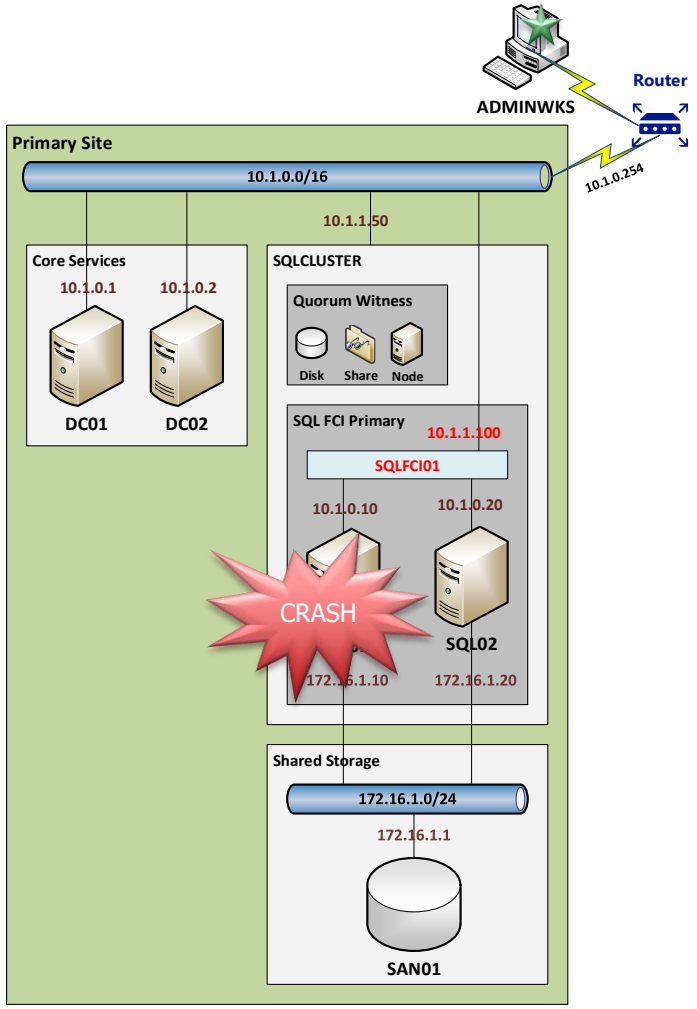
# FCI Review

- Protects an entire SQL Server instance
  - All databases including system databases (logins, jobs, etc.)
  - Automatic failover (client connects to a VNN/IP Address)
- Requires Shared Storage
- Geographic DR with SQL Server < 2012
  - Proprietary solutions for Stretch V-LAN
  - Proprietary solutions for storage replication
- Multi-subnet support >=2012

# Normal operations with AlwaysOn FCI

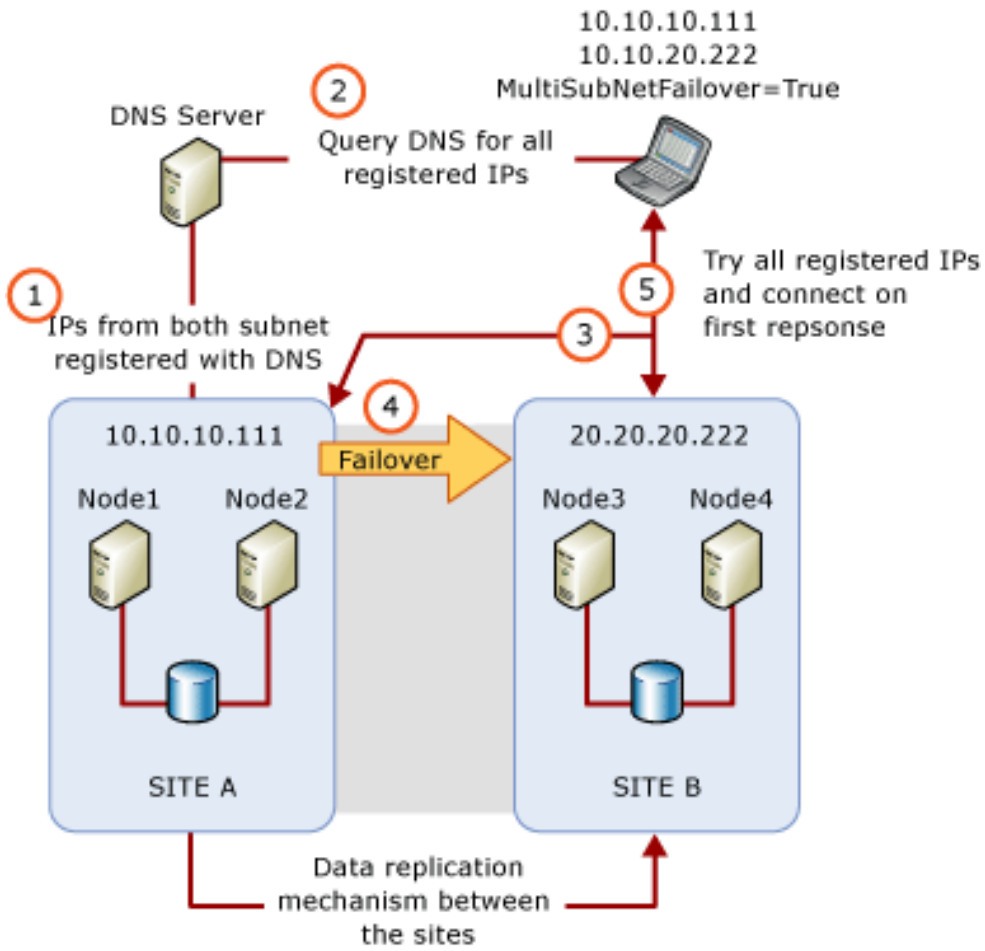


# Node unavailability with AlwaysOn FCI



# Multi-Subnet Support

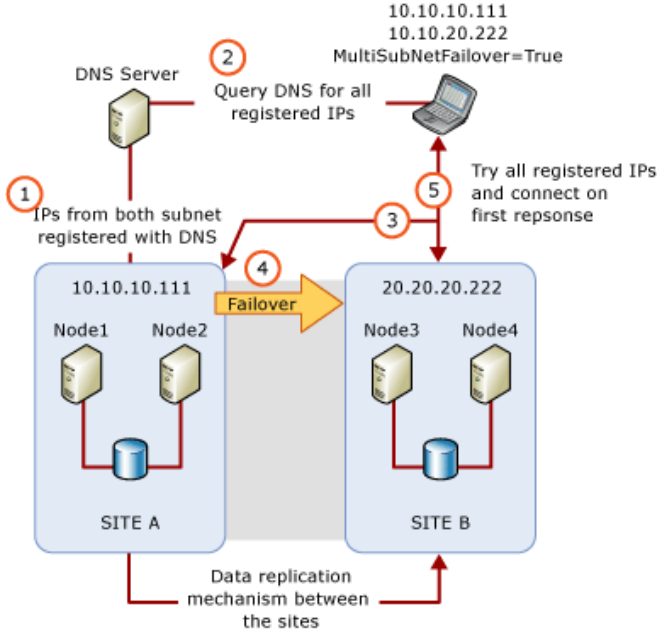
- Nodes on different subnets
  - IP Address resources in OR
- Still need storage replication
- (Enterprise Edition required)




http://msdn.microsoft.com/en-us/library/ff878716.aspx

# Multi-Subnet: how do clients reconnect?

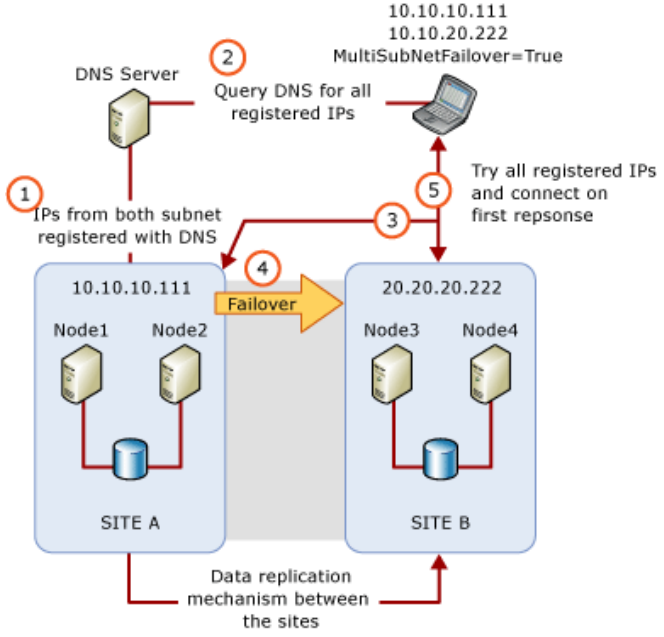
- New clients
  - Try IP addresses in order
  - Avoid DNS update latencies
  - New connection string parameter
    - MultiSubnetFailover=True
    - Try IP addresses in parallel
- Old clients can be problematic…
  - Check driver support!
  - … OleDB Provider fixed 30 March 2018!!
    - https://blogs.msdn.microsoft.com/sqlnativeclient/2017/10/06/announcing-the-new-release-of-ole-db-driver-for-sql-server
    - https://blogs.msdn.microsoft.com/sqlreleaseservices/released-microsoft-ole-db-driver-for-sql-server




http://msdn.microsoft.com/en-us/library/ff878716.aspx

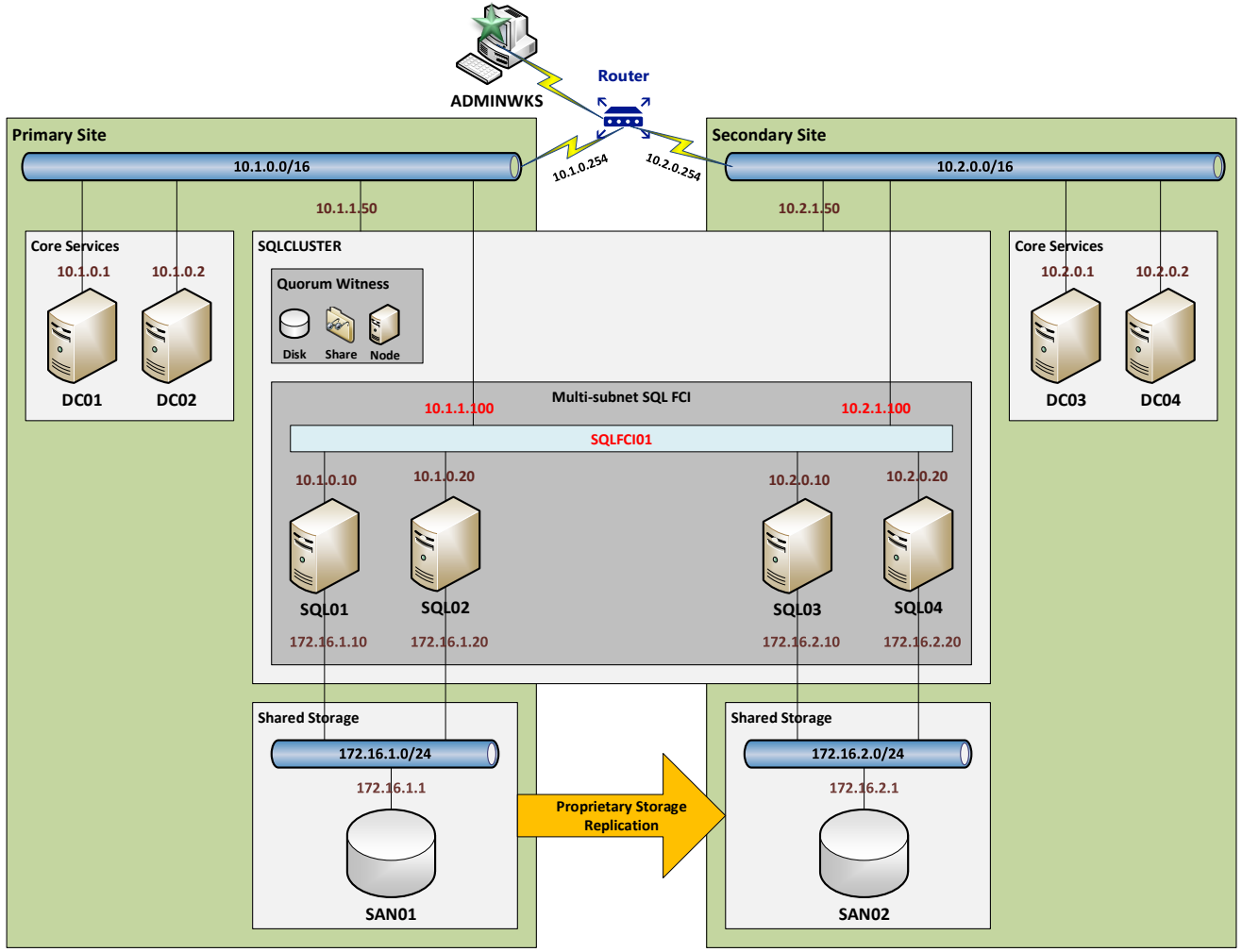
# Multi-Subnet problem with old clients

- Applies to
  - Libraries without support for MultiSubnetFailover
  - Connection strings that can't be modified
- Origin
  - Multiple IP registration in DNS
  - Library timeout < TCP timeout (15 vs 21 sec.)
- Mitigation
  - Raise library timeout > 30-40 sec. (if possible)
  - Avoid registering all IPs (affects everyone)
    - Reduce DNS updates latency (HostRecordTTL)
    - Reduce DNS cache expiration on the client
  - For AG: implement both on separate Availability Groups




http://msdn.microsoft.com/en-us/library/ff878716.aspx

# Normal operation with multi-subnet

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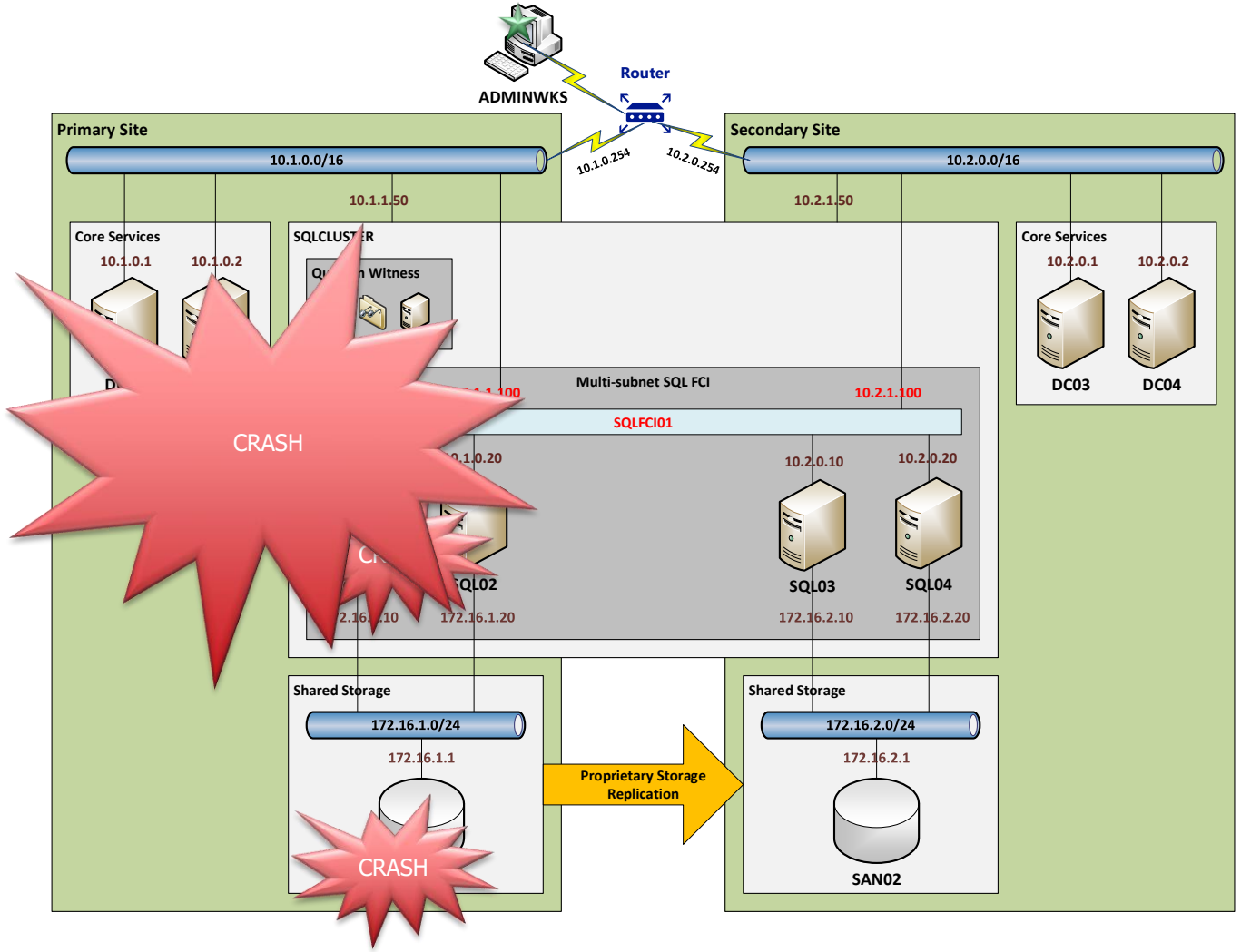
# AlwaysOn FCI multi-subnet resilience

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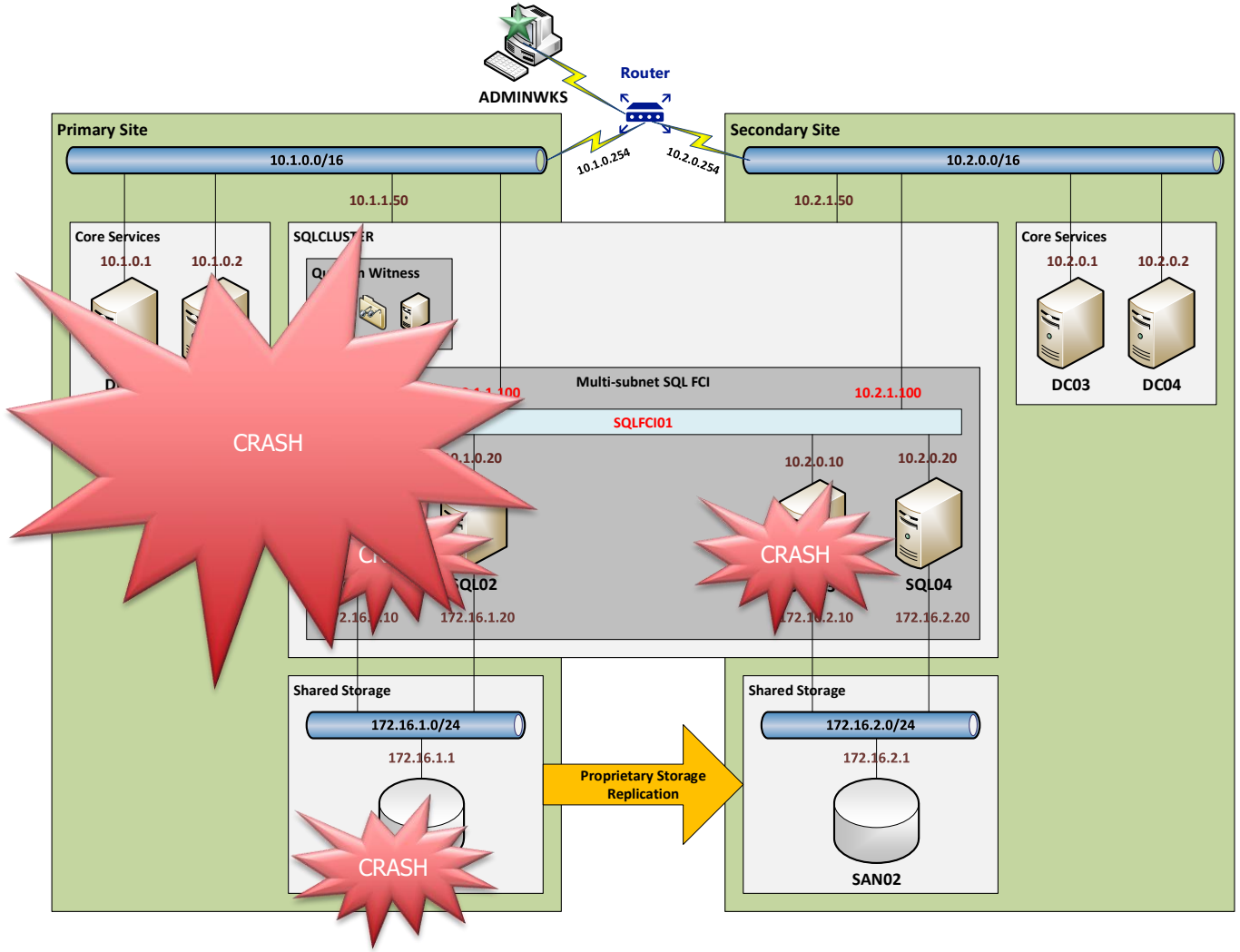
# AlwaysOn FCI multi-subnet DR with HA

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# AlwaysOn FCI in SQL Server 2014

- Support for Windows Server 2012 CSV
- Support for FCI in Sysprep
- Same support as for Availability Groups in
  - `sys.dm_hadr_cluster`
  - `sys.dm_hadr_cluster_members`
  - `sys.dm_hadr_cluster_networks`
- New DMV
  - `sys.dm_io_cluster_valid_path_names`

# AlwaysOn FCI in SQL Server 2016

- Group Managed Service Accounts (gMSA)
  - Managed directly by AD
  - Automatic password rotation
  - SQL Server 2014 Supported on Windows Server 2012 R2 + hotfix

# AlwaysOn FCI in SQL Server 2017

- Support for Linux

# AlwaysOn AG

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# Data replicas review

- Log Shipping
- Database Mirroring
- Replication
- SQL Server 2012+: Availability Groups

# Log Shipping

- Advantages
  - Entire database replicas
  - Multiple replicas per database
  - Simple and robust
- Disadvantages
  - Only asynchronous replicas
  - Readable replicas with trade-offs
    - Updated to last log restore
    - Need to disconnect clients before restoring

# Database Mirroring

- Advantages
  - Entire database replicas
  - Synchronous and asynchronous (EE) replicas
  - Simple and robust
- Disadvantages
  - Only one replica per database
  - Readable replicas need Database Snapshot
    - Enterprise Edition
    - Updated to last snapshot creation time
    - Need to disconnect or handle multiple snapshot names

# Replication

- Advantages
  - More granular (depends on scenario)
  - Multiple replicas
  - Readable & updateable replicas
- Disadvantages
  - More granular (depends on scenario)
  - Only asynchronous replicas
  - Typically higher latency
  - Complexity and maintenance overhead

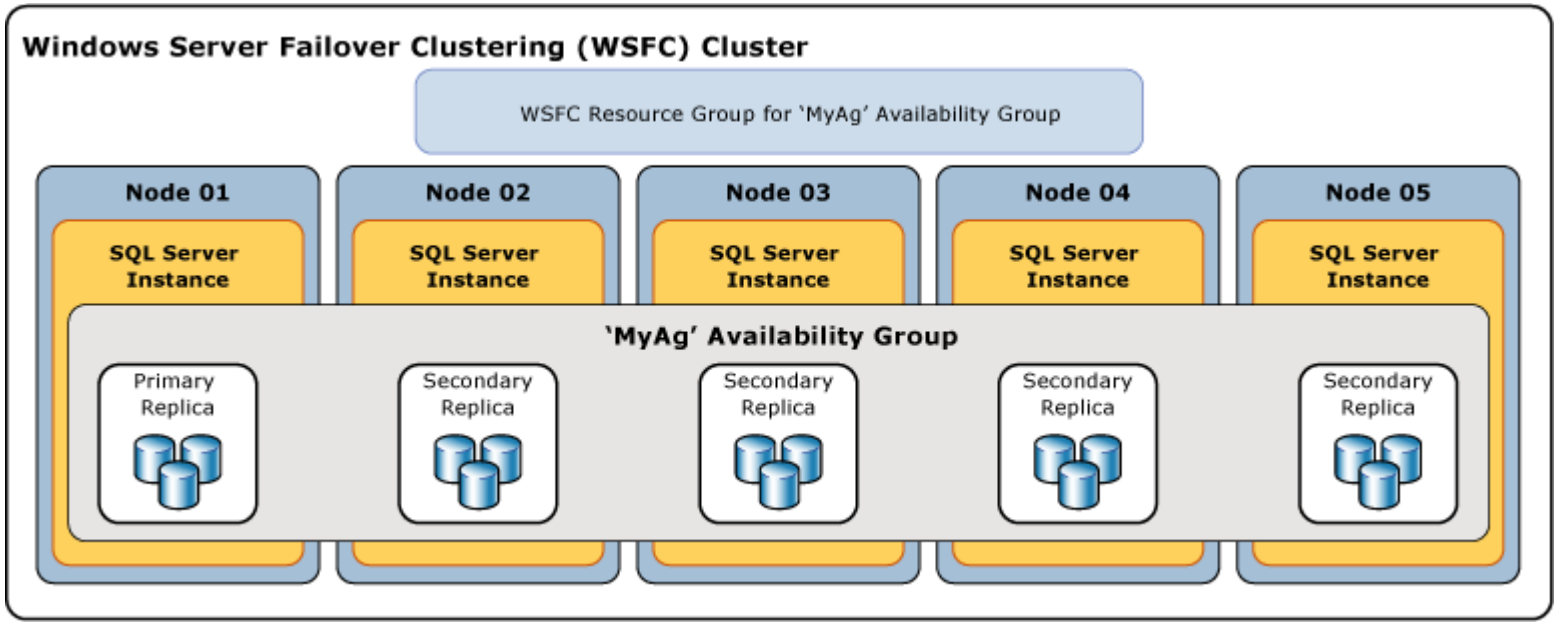
# Common Problems

- Manual Failover*
- No coordination to failover multiple DBs
- No backups on replicas*
- Specific limitations
  - E.g. FILESTREAM, cross-database consistency of transactions, only one replica

# AG as evolution of Database Mirroring

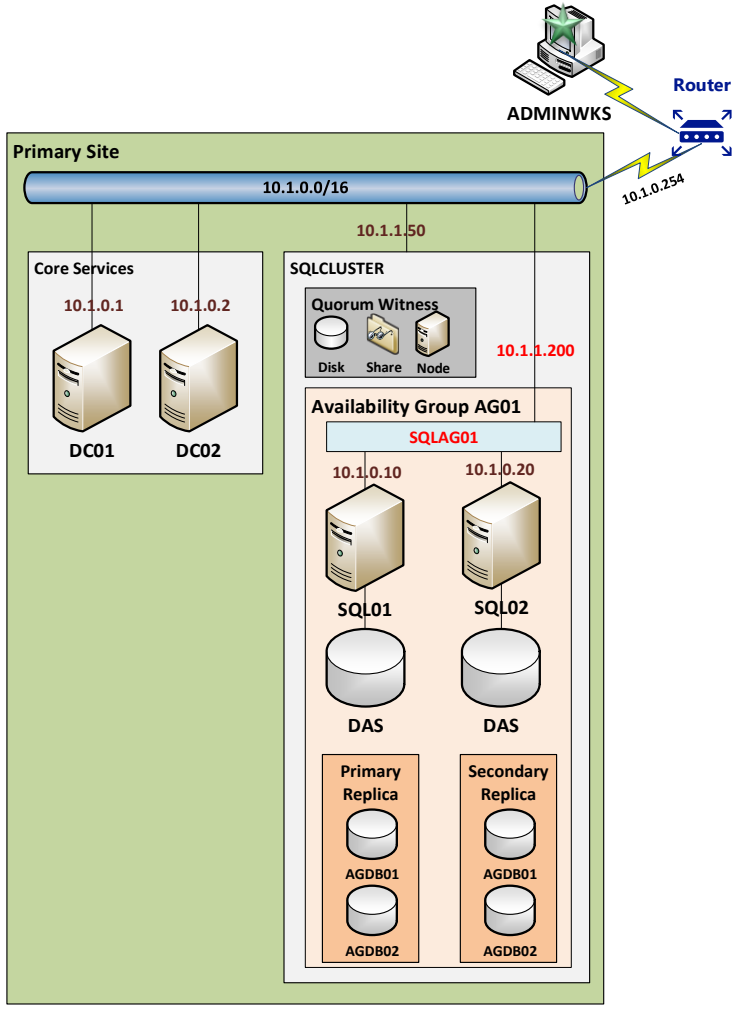
- Coordinated failover of multiple DBs
- Multiple replicas (up to 4 initially)*
- Synchronous/asynchronous replicas
- Up-to-date readable replicas
- Ability to off-load maintenance on replicas
  - e.g. backups

# AG Overview

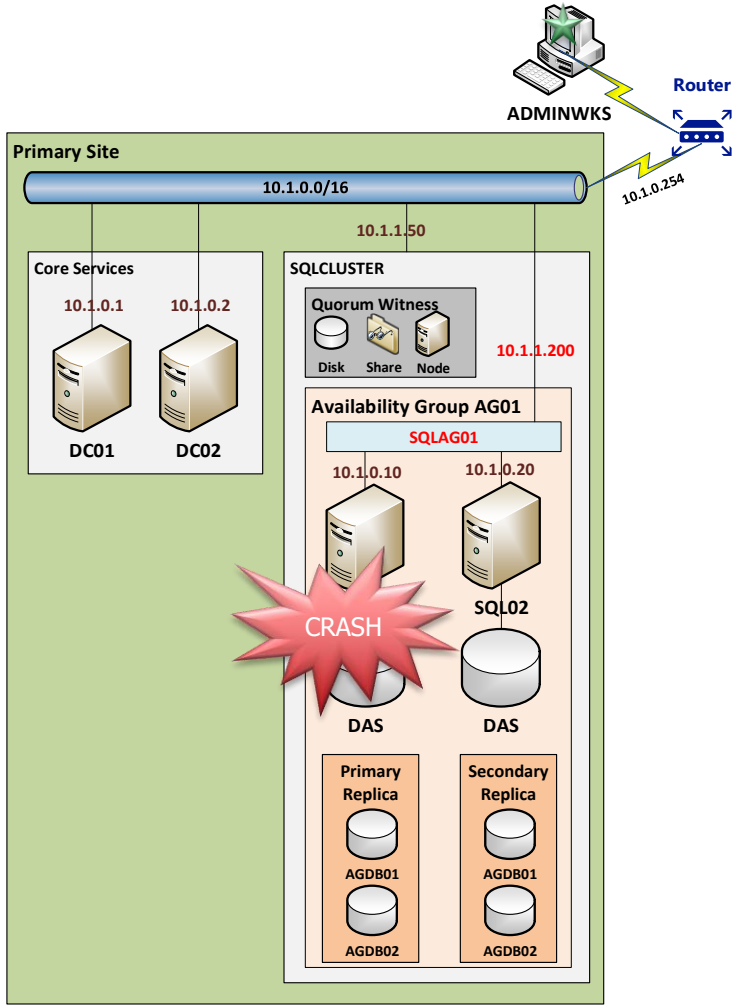



https://docs.microsoft.com/en-us/sql/database-engine/availability-groups/windows/overview-of-always-on-availability-groups-sql-server

# Normal operations with AlwaysOn AG



# Node unavailability with AlwaysOn AG

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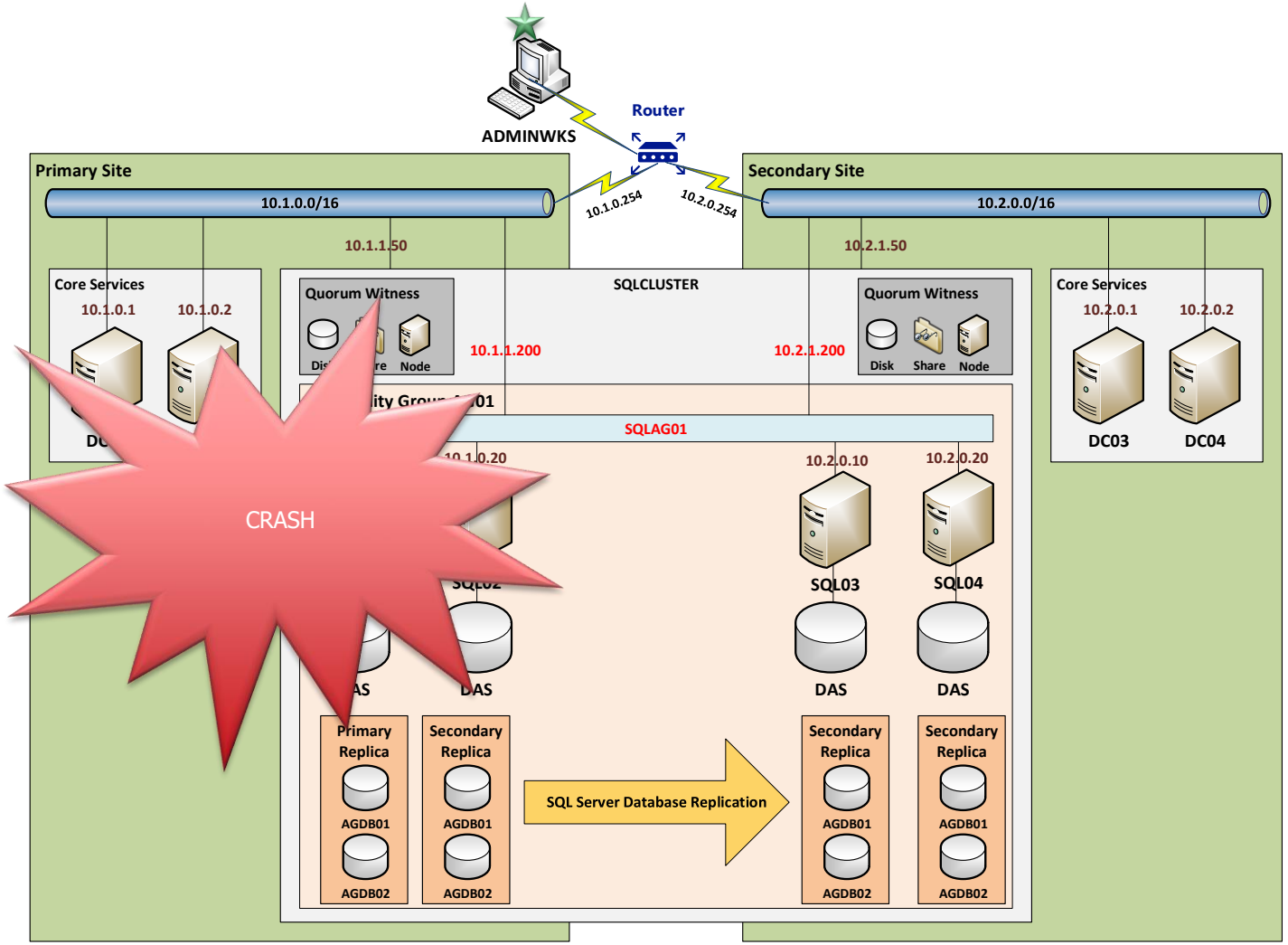
# AlwaysOn AG multi-subnet resilience

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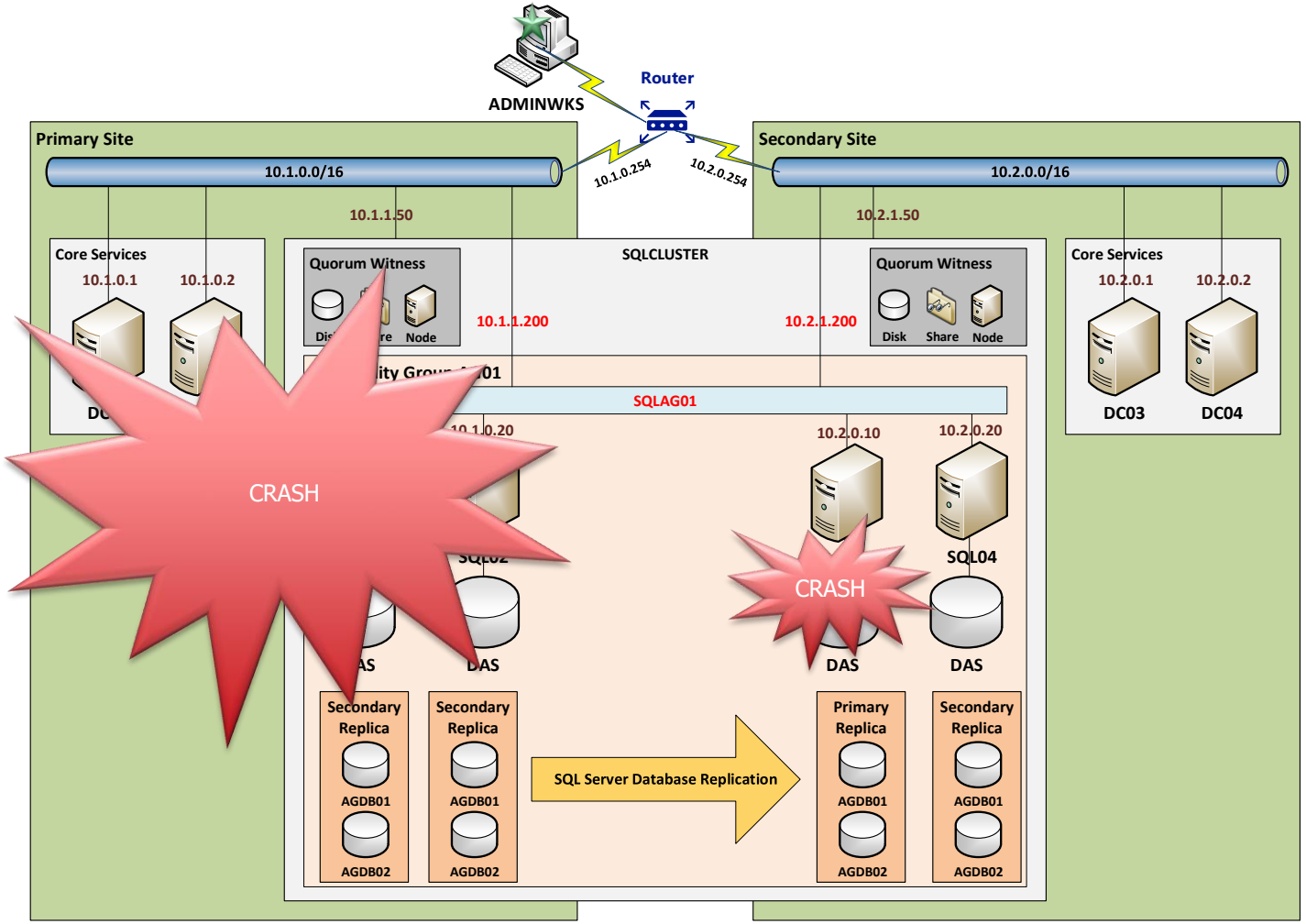
# AlwaysOn AG multi-subnet DR with HA

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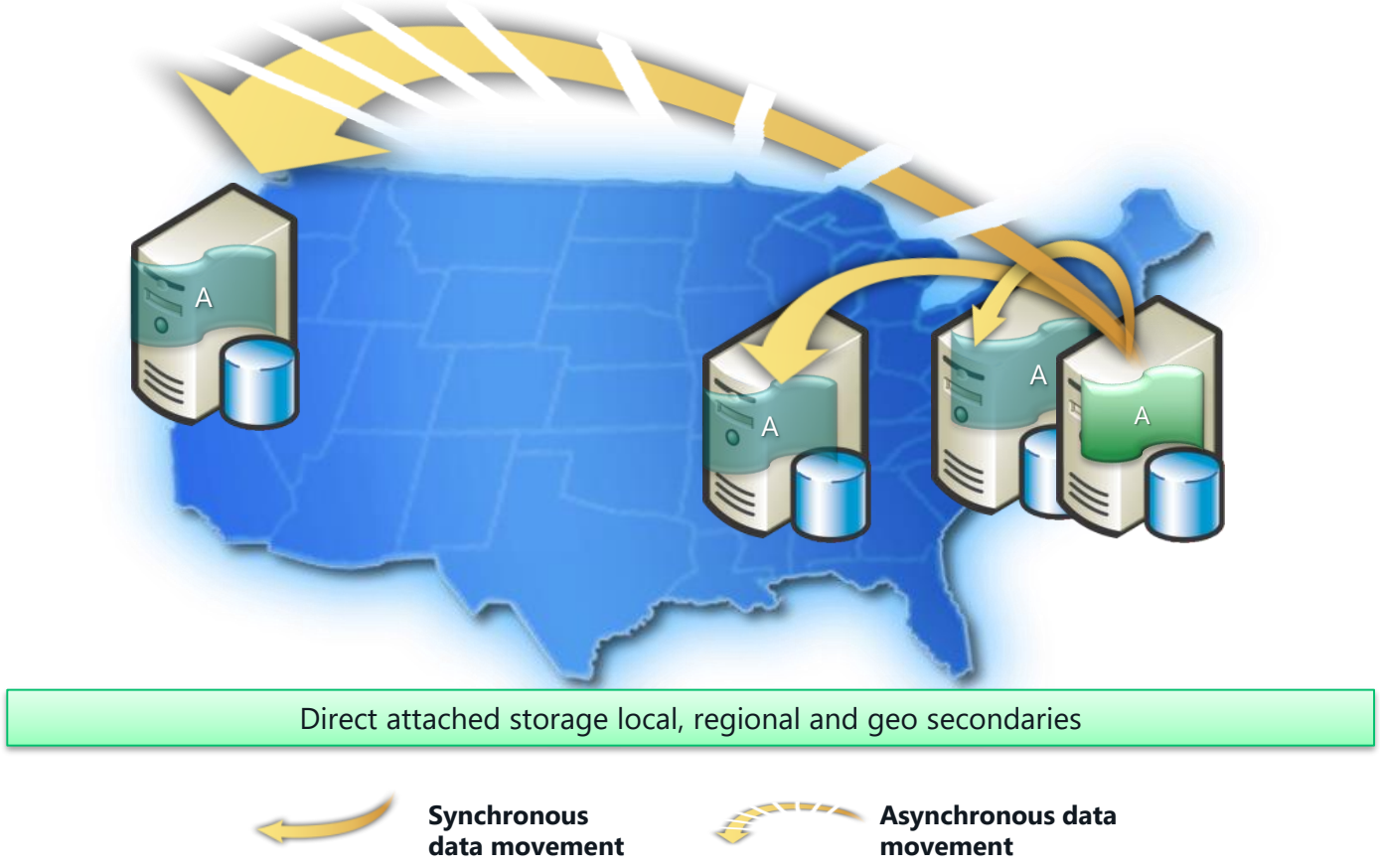
# Read-Only Access

- Replicas roles
  - Primary: **`READ_WRITE|ALL`**
  - Secondary: **`NO|READ_ONLY|ALL`**
- New connection property
  - **`ApplicationIntent`**
  - Name slightly different across client APIs
- Read-Only Routing

# AlwaysOn AG Topology Example



# Read-Only Access Capacity Planning

- Synchronization latency of replica can raise
  - because of the read I/O activity
- Impact on tempdb
  - Temporary statistics
    - Automatically created
    - Gone after failover
  - Read Committed Snapshot Isolation
    - 14 bytes added to rows at primary (on updates)

# Problem with Replicas at Database level

- Need Manual synchronization for objects outside database scope
  - New logins/users, Jobs, …
- Solution for logins/users
  - Contained databases
  - Problem with **`ApplicationIntent=ReadOnly`**

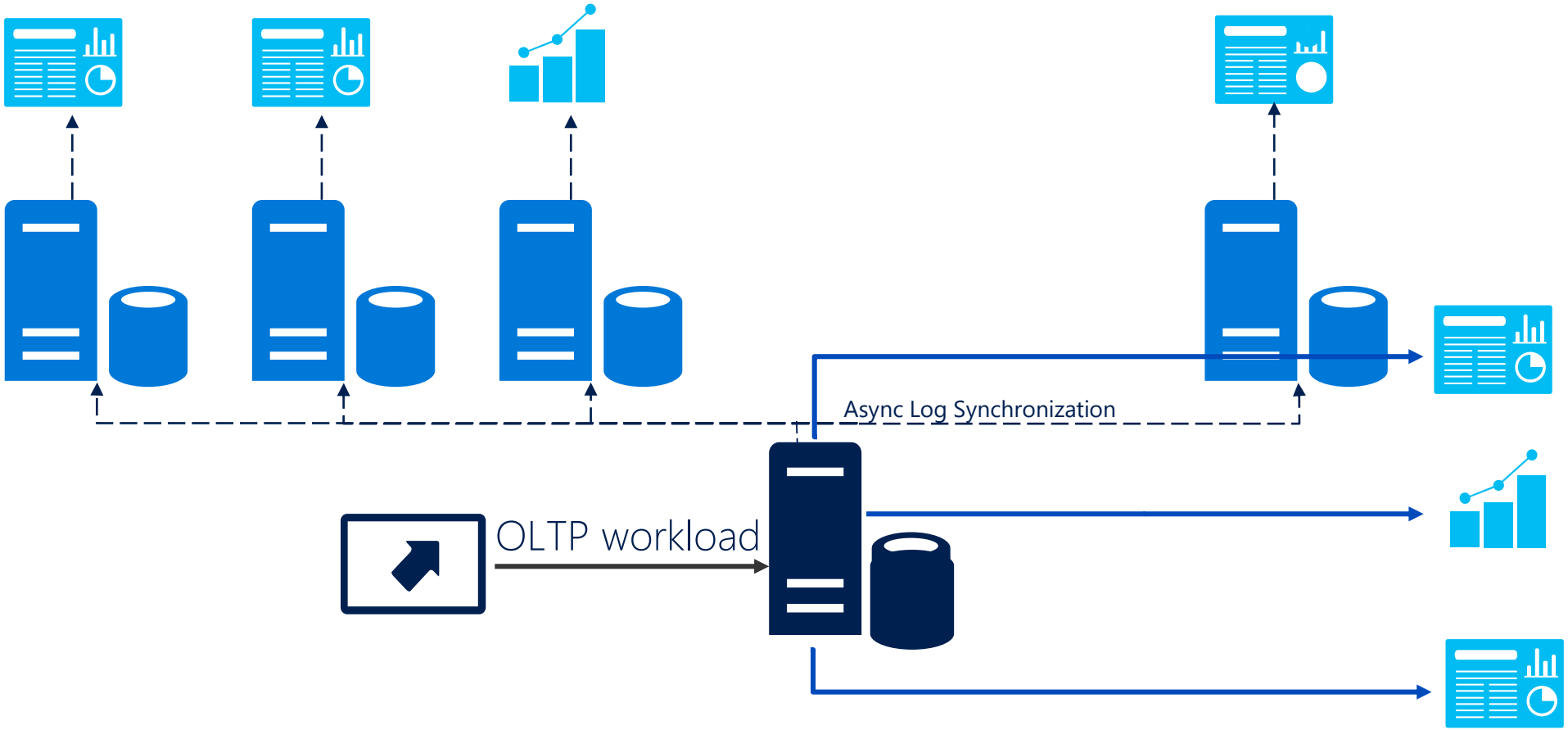
# Availability Groups in SQL Server 2014

- Up to 8 secondary replicas (was 4)
  - still only 2 can be in synchronous mode
- Secondary readable replicas remain online
  - when disconnected from primary
  - When there's a quorum loss
- New DMV
  - `sys.fn_hadr_is_primary_replica`
- Add Azure Replica Wizard

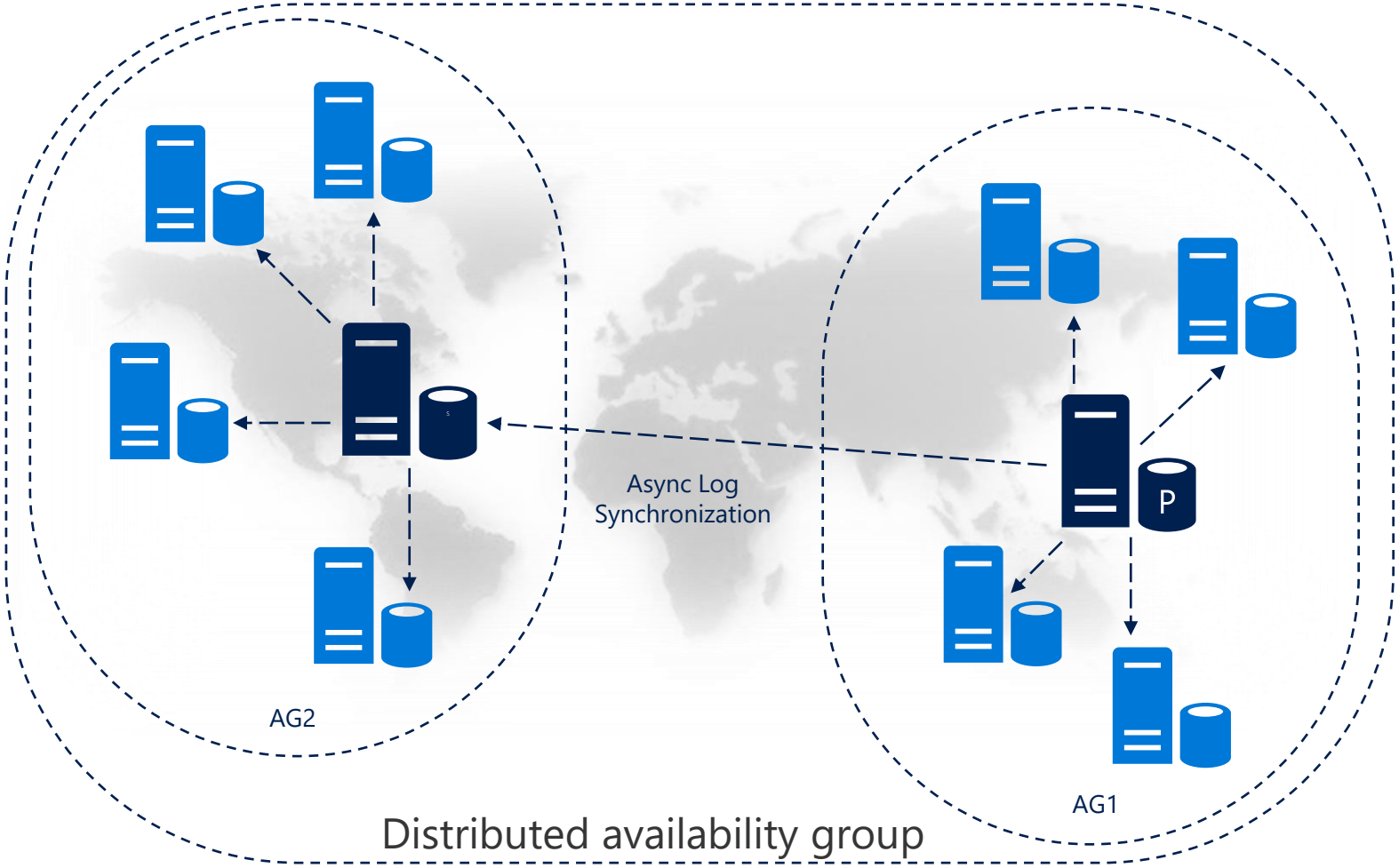
# Availability Groups in SQL Server 2016

- Up to 3 synchronous replicas with automatic failover (was 2)
- Enhanched database failover (fail on write transaction failure)
- Secondary replica seeding (instead of backup/restore)
- Read-only secondary replicas load balancing
- MSDTC supported with Windows Server 2016 (cross-instance only)
- Database encryption support (eg. SSIDB)
- Cross-domain and domain-independent AG
- Distributed AG
- Log record transport re-written to be much faster

# Scalability with Read-Only AG in SQL 2016



# Scalability with Distributed AG in SQL 2016



# Basic Availability Groups in SQL Server 2016

- Availabile with Standard Edition
  - Solution for deprecated Database Mirorring (almost…)
- Limited to 2 replicas
  - Synchronous or asynchronous (DBM required EE for async!)
  - Replica can be in Azure
- Limited to 1 database per Availability Group
- Secondary replica not accessible
  - eg. read-only, backups…

# AG Turbocharged in SQL Server 2016

- Problem
  - Performance not adequate in some synchronous replica scenarios
  - Higher scalability required with more/faster resources
  - Target: 95% performance with 1 synchronous replica
- Many optimizations in SQL Server 2016
  - Eg. reduced number of needed worker thread, use more parallelism, reduced contention
- Results
  - 95% with 1 synchronous replica, 90% with 2 replicas

# Availability Groups in SQL Server 2017

- Cross-database transaction support (MSDTC)
- Minimum number of (sync) replica to commit
- Clusterless sceanarios support
  - **`CLUSTER_TYPE = NONE`**
    - e.g. read-only scale out replicas not used for HA
- Linux support
  - **`CLUSTER_TYPE = EXTERNAL (Pacemaker)`**
  - **`CLUSTER_TYPE = NONE`**
    - Cross-OS migration Windows-Linux supported

# Advanced Scenarios

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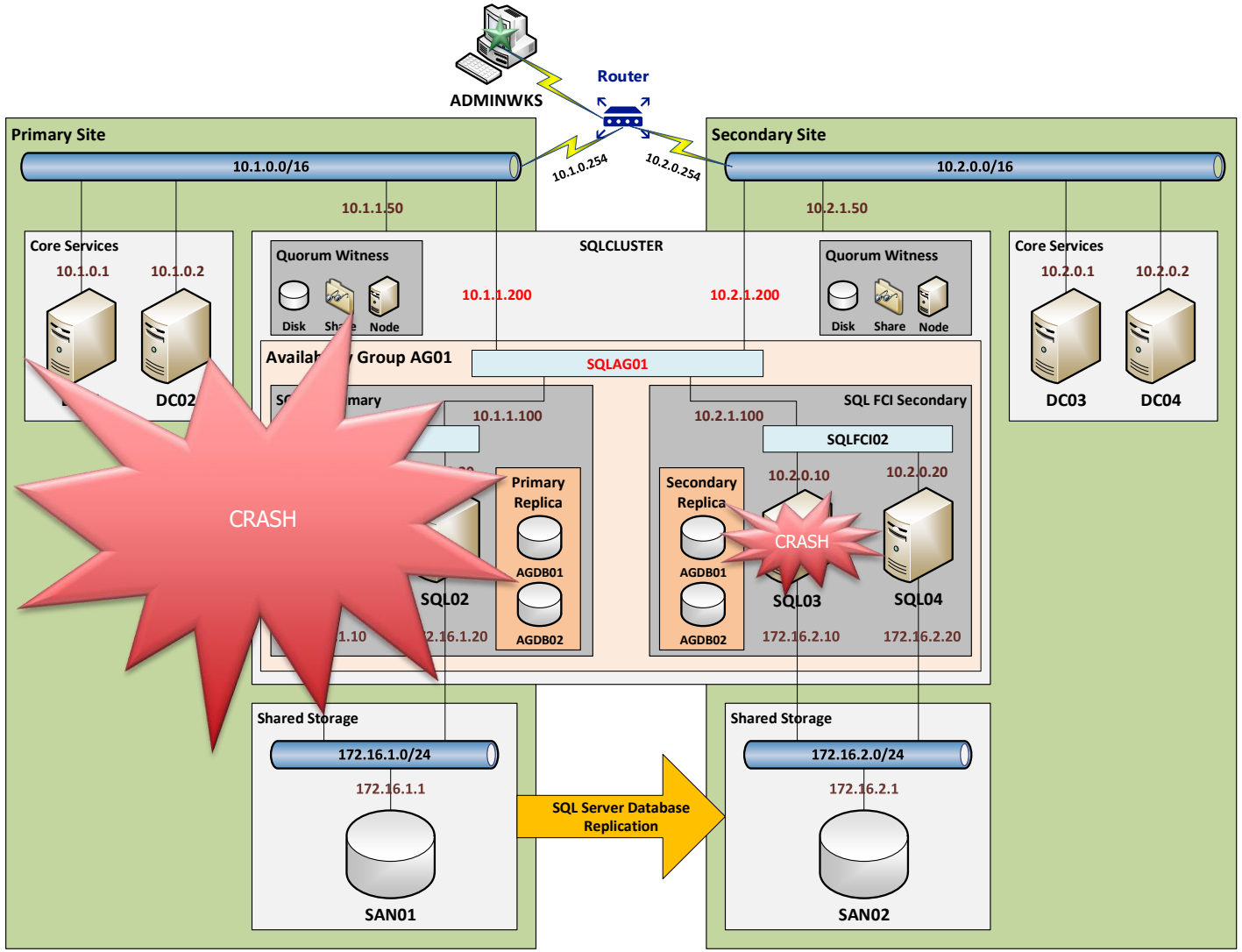
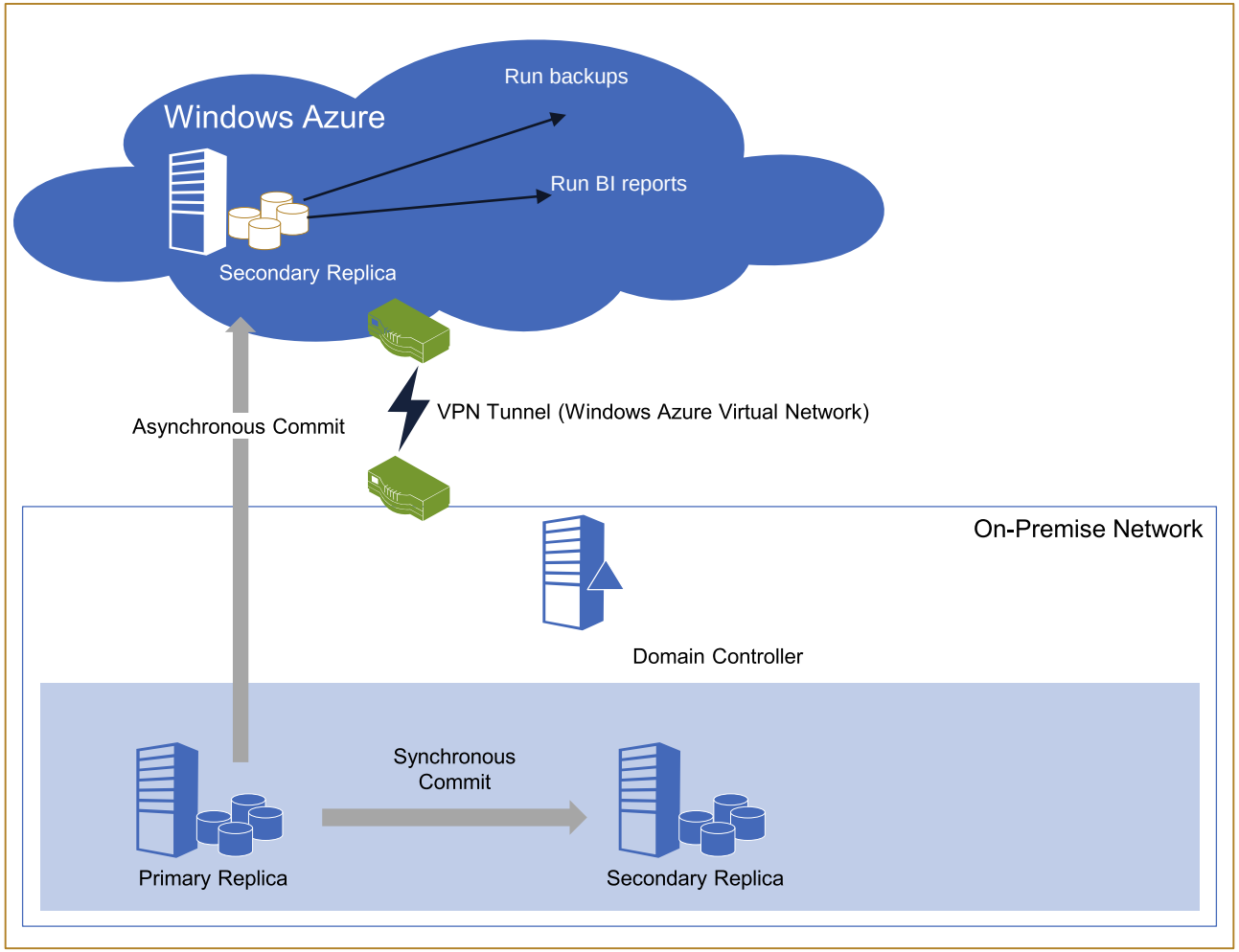
# AlwaysOn FCI+AG



# Hybrid Cloud



# Considerations

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# Other Hybrid Cloud Scenarios

- Database Mirroring
  - VPN not needed for domain authentication
  - Can use certificate based authentication

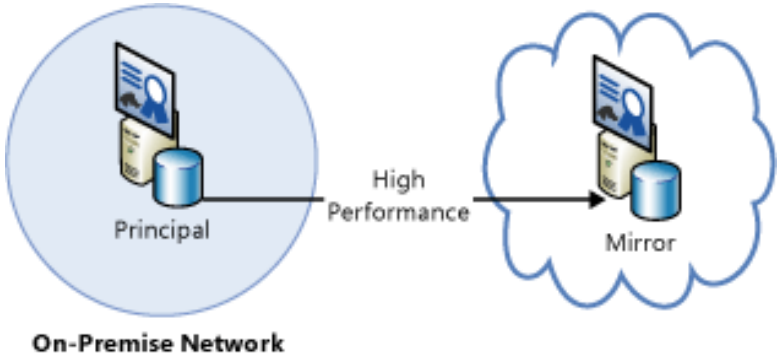


Image source: http://msdn.microsoft.com/en-us/library/jj870962.aspx

- Log Shipping

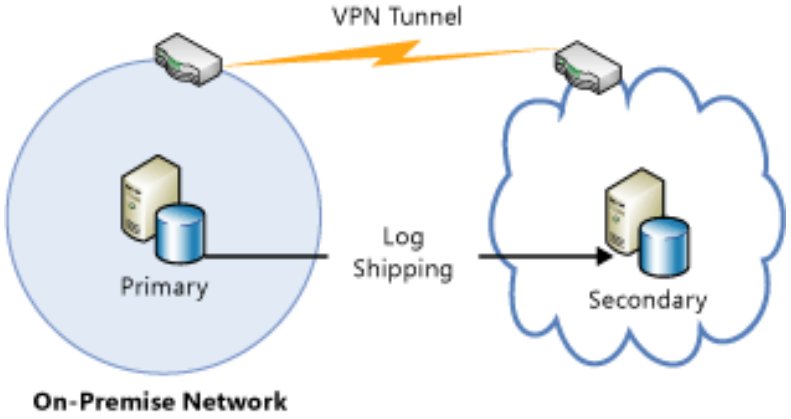


Image source: http://msdn.microsoft.com/en-us/library/jj870962.aspx

# Choice based on business requirements

- Recovery Point Objective
- Recovery Time Objective
- SLA Requirements
- Resiliency Requirements
- Deployment Span
- Read/Write Scale-out
- Cost Of Deployment
- Environmental Constraints

# «Zero downtime»?

- Excellence in designing and planning
- Excellence in execution
- Application resiliency
  - Retry logic (exponential back-off)
  - Useful in other context (deadlocks, update conflicts)

# Change in licensing

- Software Assurance needed for passive servers
  - "Beginning with SQL Server 2014, each active server licensed with SA coverage allows the installation of a single passive server used for fail-over support."
- Right to run only one free passive secondary
  - "The active server license (s) must be covered with SA, and allow for one passive secondary SQL Server, with up to the same amount of compute as the licensed active server, only."
- Impacts everything: log shipping, mirroring, AG, FCI
- SQL Server 2014-2017 Licensing Datasheets
  - http://download.microsoft.com/download/6/6/F/66FF3259-1466-4BBA-A505-2E3DA5B2B1FA/SQL_Server_2014_Licensing_Datasheet.pdf
  - http://download.microsoft.com/download/F/D/5/FD5E5C28-6973-4273-8737-D69AA3BEA243/SQL_Server_2016_Licensing_Datasheet_EN_US.pdf
  - http://download.microsoft.com/download/7/8/c/78cdf005-97c1-4129-926b-ce4a6fe92cf5/sql_server_2017_licensing_guide.pdf
- SQL Server 2017 Licensing Guide
  - http://download.microsoft.com/download/7/8/c/78cdf005-97c1-4129-926b-ce4a6fe92cf5/sql_server_2017_licensing_guide.pdf

# AG Main Resources

- Official documentation
  - https://docs.microsoft.com/en-us/sql/sql-server/failover-clusters/windows/always-on-failover-cluster-instances-sql-server
  - https://docs.microsoft.com/en-us/sql/database-engine/availability-groups/windows/always-on-availability-groups-sql-server
- Blogs
  - CSS SQL Server Engineers
    - http://blogs.msdn.com/b/psssql/archive/tags/alwayson
  - SQL AlwaysOn Blog
    - http://blogs.msdn.com/b/sqlalwayson (stale)

# AG Whitepapers series

- AlwaysOn Architecture Guide: Building a High Availability and Disaster Recovery Solution by Using AlwaysOn Availability Groups
  - http://msdn.microsoft.com/en-us/library/jj191711.aspx
- AlwaysOn Architecture Guide: Building a High Availability and Disaster Recovery Solution by Using Failover Cluster Instances and Availability Groups
  - http://msdn.microsoft.com/en-us/library/jj215886.aspx
- AlwaysOn Solution Guide: Offloading Read-Only Workloads to Secondary Replicas
  - http://msdn.microsoft.com/en-us/library/jj542414.aspx
- Cross-cluster Migration of AlwaysOn Availability Groups for Operating System Upgrades
  - http://msdn.microsoft.com/en-us/library/jj873730.aspx
- Microsoft SQL Server AlwaysOn Solutions Guide for High Availability and Disaster Recovery
  - http://msdn.microsoft.com/en-us/library/hh781257.aspx
- Migration Guide: Migrating to AlwaysOn Availability Groups from Prior Deployments Combining Database Mirroring and Log Shipping
  - http://msdn.microsoft.com/en-us/library/jj635217.aspx
- Multisite Failover Cluster Instance
  - http://sqlcat.com/sqlcat/b/whitepapers/archive/2011/12/22/sql-server-2012-alwayson_3a00_-multisite-failover-cluster-instance.aspx
- SQL Server 2012 AlwaysOn High Availability and Disaster Recovery Design Patterns
  - http://sqlcat.com/sqlcat/b/msdnmirror/archive/2011/12/22/sql-server-2012-alwayson-high-availability-and-disaster-recovery-design-patterns.aspx

# Resources SQL Server 2019

- Ignite 2018 Video
  - Architecting a highly available database platform with SQL Server
    - https://www.youtube.com/watch?v=5h1Xkh9CU-c

# Thanks!

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